

Experiment 5

Aim : Write a program to simulate Mutual Exclusion (Raymond's Tree-Based Algorithm).

Theory :

- Mutual exclusion ensures that only one process ~~as~~ accesses a shared resources at a time in a distributed system.
- Raymond's algorithm is a distributed, tree-based approach that efficiently grants access to a shared resources while minimizing message overhead.
- Steps :

1) Requesting the token

- ⇒ A process that needs the token checks if it already holds it :-
- IF yes → It enters the critical section
 - IF no → It forwards request to its parent node in the tree.

2) Token passing

- ⇒ The request propagates upward until it reaches the current token holder.
- The token holder sends the token downward through the shortest path to the requesting process.

3) Entering the critical section

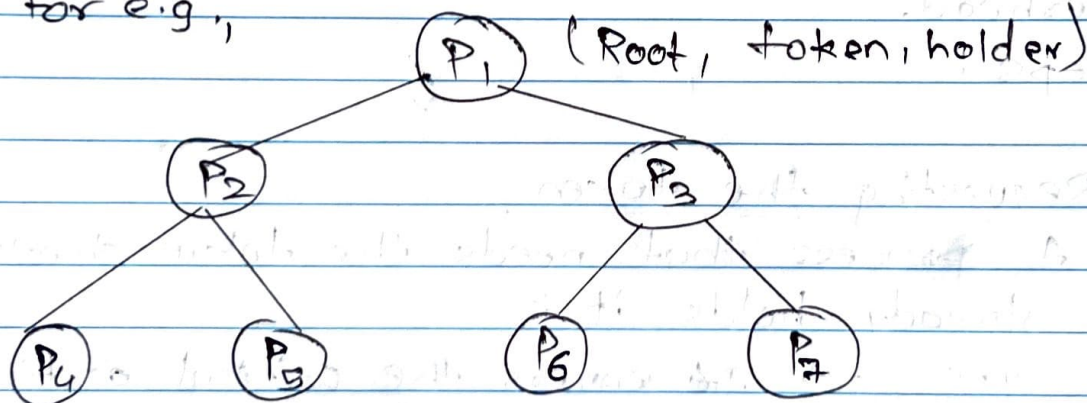
⇒ When a process receives the token, it removes itself from the queue and enters the critical section.

4) Releasing the token

⇒ After execution, the process checks its queue:

- If queue is empty → It keeps the token.
- If queue is not empty → It passes the token to the next requester in the queue.

- For e.g.,



Scenario: P₇ Requests the token

- 1) P₇ → P₃ (Inced the token)
- 2) P₃ → P₁ (P₇ needs the token)
- 3) P₁ (Token holder) → P₃ (Passes token)
- 4) P₃ → P₇ (Token reaches P₇)
- 5) P₇ enters the critical section

(A)

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